

DETAILED DESCRIPTION OF THE INVENTION

Executive Summary of Inventive Architecture

The present disclosure describes coordinated systems and methods for generating a user-specific penile anatomical geometry profile through synchronized spatial acquisition and computational reconstruction. The disclosed architecture integrates radial distance sensing, longitudinal displacement indexing, geometric reconstruction modeling, encoded profile generation, secure encryption frameworks, and modular interoperability pathways.

At its core, the invention establishes a structured methodology for transforming synchronized depth measurements and positional data into a standardized digital anatomical geometry representation. Unlike conventional scalar measurement approaches, the disclosed system produces a reproducible, indexed, and computationally usable geometry profile capable of supporting calibration, longitudinal tracking, compatibility modeling, and multi-node biometric integration.

The architecture supports multiple sensor configurations, including single-sensor, multi-sensor, and multi-zone depth modules, in combination with optical motion sensing, inertial stabilization, physiological validation mechanisms, and adaptive sampling strategies. Reconstruction techniques may employ deterministic modeling, interpolation frameworks, machine learning–assisted estimation, and constraint-based surface continuity enforcement.

The resulting geometry profile may be encoded, encrypted, normalized, abstracted, tokenized, or transformed into privacy-preserving structural signatures suitable for secure storage, distributed processing, compatibility scoring, or system-level integration. The architecture further accommodates consumer, research, and clinical deployment configurations without altering the foundational synchronized acquisition methodology.

Collectively, the disclosed embodiments establish a scalable, modular, and future-adaptable platform for penile anatomical geometry profiling capable of supporting standalone deployment or integration within broader biometric ecosystems. The written description provided herein supplies a comprehensive foundation for subsequent non-provisional claim drafting encompassing device, system, and method implementations.